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Reliability Growth During Flight Test

20 July 2013

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Introduction

This is a simple analysis of the process of finding and fixing defects in flight hardware reprising some of the results in the reference. It is intended to support management decisions on the extent of development analysis and testing, of the scope of quality assurance, and of the depth of the FMEA process and related test instrumentation. Although it's easy to derive more comprehensive models, they all have more parameters to estimate a priori, making them , in the end, no more useful.

Analysis

In the narrow technical sense, a defect or fault or failure not in compliance with the applicable system spec. A fault becomes a failure when it is realized in flight. Assume that occurrence of latent design / workmanship defects is governed by binomial statistics with the occurrence of each fault independent of all others. In the present context, fault detection happens when a fault is realized, detected and corrected before the next flight. The analysis below is done on an expectation basis.

Let

D_n = Number of lethal latent defects after n flight tests,
p = Probability that any unique latent defect will occur during a specific flight,
T = Number of flight tests,
d = Probability that a lethal latent defect will be detected, characterized and corrected if it is manifested during a specific flight test, and
R = Reliability.

First, note that reliability is the probability that no faults occur in a flight\. For the very first launch,

$$\mathbf{R = (1 - p)^{Do}}$$

In general, reliability = Prob (no defects are manifested during a flight after n flight tests)

$$= \mathbf{(1 - p)^{Dn}}$$

$$= \mathbf{(1 - p)^{Do}} \text{ for the first flight test.}$$

The expected number of defects manifested in the **n**th flight (after **n – 1** tests) is

= **$D_{n-1}\, p$**

And, the expected number of defects corrected as a result of the **n**th flight test is

= **$D_{n-1}\, p\, d$**

Then the expected number of latent lethal defects remaining after the **n**th flight is

$D_n = D_{n-1} - D_{n-1}\, p\, d = D_{n-1}\,(1 - p\, d)$

Thus, **$D_1 = D_o\,(1 - p\, d)$,** and

$D_2 = D_1\,(1 - p\, d) = D_o\,(1 - p\, d)^2$

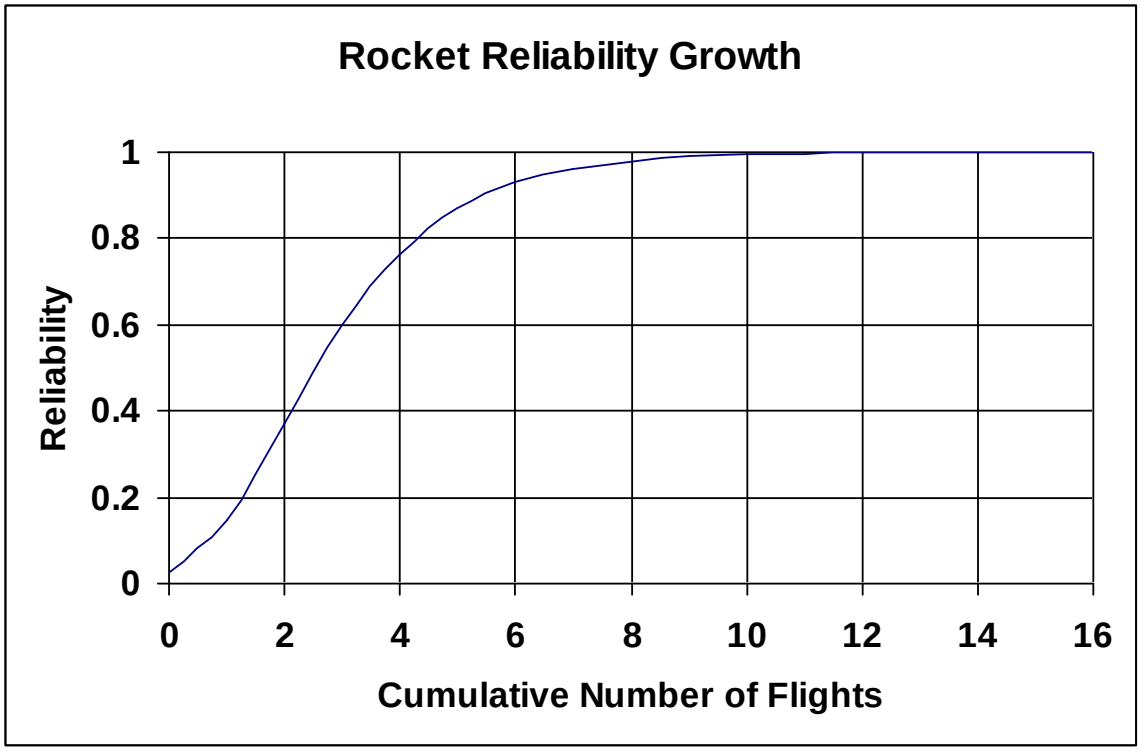
In general, after **n** flight tests,

$D_n = D_o\,(1 - p\, d)^n$

Finally, the reliability is given by $\boxed{R = (1 - p)^{D_o (1 - p\, d)^T}}$

Results

This analysis can be readily coded into Excel™ to generate RELGRWTH.xls. Typical results are shown below for $D_0 = 6$, $p = 0.6$ and $d = 0.8$. The reference considered a set of development flight test histories and found these parameters to be representative.



Reference
W. Schaechter and C. P. Hoult, "Economic Considerations in Sounding Rocket Utilization", Proceedings of the Sounding Rocket Vehicle Technology Specialist Conference, Williamsburg, VA, February 1967.